Algebra, Dummit and Foote - Chapter 13: Field Theory

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March 25, 2026 $\sim$ May 15, 2026

§2 Algebraic Extensions

8.

Since the minimal polynomials for $\sqrt{D_1}$ and $\sqrt{D_2}$ are both degree 2, it is clear that regardless of whether $\sqrt{D_1 D_2}$ is a square, $F(\sqrt{D_1}, \sqrt{D_2})$ is a degree 2 or degree 4 extension over F. Suppose $D_1 D_2 = a^2$ for some $a \in F$. Since D_1 is not a square in F, $F(\sqrt{D_1})$ is a degree 2 extension over F. Then $\sqrt{D_2}$ satisfies the polynomial $x^2 - D_2$, which can be manipulated as

$$\begin{aligned} x^2 - D_2 &= 0 \\ D_1 x^2 - D_1 D_2 &= 0 \\ x^2 &= \frac{a^2}{D_1} \\ x &= \frac{a}{\sqrt{D_1}} \end{aligned}$$

hence $\sqrt{D_2} \in F(\sqrt{D_1})$, which shows that $F(\sqrt{D_1}, \sqrt{D_2})$ is a degree 2 extension over F.

Conversely, suppose $F(\sqrt{D_1}, \sqrt{D_2})$ is a degree 2 extension over F. Then necessarily, $\sqrt{D_2} \in F(\sqrt{D_1})$, which means

$$\sqrt{D_2} = u + v\sqrt{D_1} \qquad (u, v \in F). \tag{1}$$

Square both sides to obtain

$$D_2 = u^2 + 2uv\sqrt{D_1} + v^2 D_1. \tag{2}$$

Since $\sqrt{D_1} \notin F$, (2) forces $u = 0$ or $v = 0$. By the same reason, (1) forces $v \neq 0$, so $u = 0$. Hence multiplying both sides of (2) by D_1 gives

$$D_1 D_2 = v^2 D_1^2,$$

i.e., $D_1 D_2$ is a square in F, as was to be shown. The contrapositive of this result says that if $D_1 D_2$ is not a square, then $F(\sqrt{D_1}, \sqrt{D_2})$ is a degree 4 extension over F.

9.

Suppose $\sqrt{a + \sqrt{b}} = \sqrt{m} + \sqrt{n}$ for $m, n \in F$. Then $a + \sqrt{b} = m + n + 2\sqrt{mn}$ in $F(\sqrt{b})$. Since $1, \sqrt{b}$ is a basis for $F(\sqrt{b})$ over F and $\sqrt{b} \notin F$,

$$\begin{aligned} 4mn &= b \\ m + n &= a \end{aligned}$$

where the first equation is from matching $2\sqrt{mn} = \sqrt{b}$. Substituting these equations for a and b yields

$$a^2 - b = m^2 - 2mn + n^2 = (m-n)^2,$$

so $a^2 - b$ is a square in F.

Conversely, suppose $a^2 - b = c^2$ for some $c \in F$. As in the forward direction, $m, n \in F$ satisfying the system $4mn = b$ and $m + n = a$ must be found. Solving this system for n yields the polynomial

$$4n^2 - 4an + b = 0$$

in n. Since $\text{char} F \neq 2$, the quadratic equation applies to give

$$\begin{aligned} n &= \frac{4a + \sqrt{16a^2 - 16b}}{8} \\ &= \frac{4a + 4c}{8} \\ &= \frac{a+c}{2}. \end{aligned}$$

It follows $m = \frac{a-c}{2}$. By definition, such $m, n \in F$ satisfy $a + \sqrt{b} = m + n + 2\sqrt{mn}$, as desired.

If $a^2 - b$ is not a square, the result proven above says that $F(\sqrt{a + \sqrt{b}})$ cannot be a degree 2 or a degree 4 extension. If $a^2 - b$ is a square, then $F(\sqrt{a + \sqrt{b}}) = F(\sqrt{m} + \sqrt{n}) = F(\sqrt{m}, \sqrt{n})$, where the second equality follows by the following observation: $F(\sqrt{m} + \sqrt{n}) \subset F(\sqrt{m}, \sqrt{n})$ is obvious. The other inclusion follows because $\frac{m-n}{\sqrt{m}+\sqrt{n}} = \sqrt{m} - \sqrt{n}$, where the left side is clearly in $F(\sqrt{m} + \sqrt{n})$. By 8), $F(\sqrt{m}, \sqrt{n})$ is biquadratic if and only if m, n, and mn are not squares in F. But since $4mn = b$ and b is not a square by assumption, mn is not a square. To conclude, $F(\sqrt{a + \sqrt{b}})$ is biquadratic if and only if $a^2 - b$ is a square and one of m or n is not a square.

14.

$F(\alpha^2) \subset F(\alpha)$ automatically. If $F(\alpha) \subsetneq F(\alpha^2)$, then $F(\alpha)$ is a nontrivial extension of $F(\alpha^2)$. α satisfies $x^2 - \alpha^2$ over $F(\alpha^2)$, and this is the minimal polynomial for α since the degree of α in $F(\alpha^2)$ must be greater than 1. $[F(\alpha) : F] = [F(\alpha) : F(\alpha^2)][F(\alpha^2) : F]$ says that $2 \mid [F(\alpha) : F]$, which contradicts the assumption that the degree of α over F is odd.

15.

Suppose the claim is not true. Let $f(x)$ be minimal with respect to being an irreducible, odd-degree polynomial such that some root α of $f(x)$ gives an extension $F(\alpha)$ of F that is not formally real. Let $d := \deg \alpha = \deg f$. Since $F(\alpha) \cong F[x]/\langle f(x) \rangle$, there exist $h_1(x), \ldots, h_t(x) \in F[x]$ with $\deg h_i < d$ such that

$$-1 = h_1(\alpha)^2 + \ldots h_t(\alpha)^2$$

in $F(\alpha)$. Then there is $g(x) \in F[x]$ such that

$$-1 + f(x)g(x) = h_1(x)^2 + \ldots h_t(x)^2 \tag{3}$$

in $F[x]$. $\deg h_i < d \implies \deg h_i^2 < 2d \implies \deg fg < 2d \implies \deg g < d$. Furthermore, $\deg g$ is odd: without loss of geneeralit, suppose $h_1, \ldots, h_k$ have maximal degree amongst the h_i's. Denote their leading terms by c_i, respectively. Then $\sum_{i=1}^k lt(p_i(x)^2) = \sum_{i=1}^k (lt(p_i(x)))^2 = \sum_{i=1}^k c_i^2$, and this sum can't be 0: if it was,

$$c_2^2 + \ldots + c_k^2 = -c_1^2 \implies (c_2c_1^{-1})^2 + \ldots + (c_kc_1^{-1})^2 = -1,$$

which contradicts the assumption that F is formally real. So the degree of the right side of (3), $\deg[h_1(x)^2 + \ldots + h_t(x)^2]$, is even. That $\deg f$ is odd forces $\deg g$ even. It follows that g has an irreducible factor of odd degree, say $p(x)$. Let β be a root of p. Then in $F(\beta)$, (3) is equivalent to

$$-1 = h_1(\beta)^2 + \ldots + h_t(\beta)^2,$$

hence $F(\beta)$, given by $p(x)$, is not formally real. But $p(x)$ is an odd-degree polynomial of a lower degree than $f(x)$, contradicting the minimality of f.

16.

Take $\alpha \in R$. α is algebraic over F, so let $f(x) \in F[x]$ be the minimal polynomial for α over F. Then the homomorphism

$$\varphi : F[x] \to R;\ g(x) \mapsto g(\alpha)$$

is a well defined ring homomorphism with $\ker \varphi = \langle f(x) \rangle$, which means $F(\alpha) \subset R$. Hence $\alpha^{-1} \in R$, which shows that R is a field.

§4 Splitting Fields and Algebraic Closures

1.

$$x^4 - 2 = (x^2 - \sqrt{2})(x^2 + \sqrt{2}) = (x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - i\sqrt[4]{2})(x + i\sqrt[4]{2})$$

Gives the complete factorization of $x^4 - 2$. The dumb answer to the question is that the splitting field for $x^4 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[4]{2}, \sqrt[4]{2})$ and that its degree is 8 because $x^2 + \sqrt{2}$ is irreducible (has no root) in $\mathbb{Q}(\sqrt[4]{2})$, which can be done by realizing $\mathbb{Q}(\sqrt[4]{2})$ as $\{a + b\sqrt[4]{2} + c\sqrt{2} + d\sqrt[4]{8} \mid a, b, c, d \in \mathbb{Q}\}$.

The much easier solution is that $\mathbb{Q}(\sqrt[4]{2}, i)$ is an isomorphic splitting field for $x^4 - 2$ over $\mathbb{Q}$. Clearly $[\mathbb{Q}(\sqrt[4]{2}, i) : \mathbb{Q}(\sqrt[4]{2})] = 2$ since $\mathbb{Q}(\sqrt[4]{2})$ is a subfield of $\mathbb{R}$, whereas $i \in \mathbb{C} \setminus \mathbb{R}$. It follows that $\mathbb{Q}(\sqrt[4]{2}, i)$ is a degree 8 extension of $\mathbb{Q}$.

§5 Separable and Inseparable Extensions

4.

Let $f(x) = x^p - x + a$ and let α be a root of f. Then

$$f(\alpha + 1) = \alpha^p + 1 - (\alpha + 1) + a = f(\alpha) = 0,$$

so $\alpha + 1$ is a root of f as well. Repeating the process gives

$$\alpha, \alpha + 1, \ldots, \alpha + (p - 1)$$

as the distinct roots of f (i.e., the roots of f is precisely the set above). This shows that f is separable.

Let $p(x)$ be the minimal polynomial for α over $\mathbb{F}_p$. Then $f(x) = p(x)f_1(x)$. If $f_1 = 1$, then $f = p$ is irreducible. Suppose $f_1 \neq 1$ and let $\alpha + j_1$ be a root of f_1. Notice $p(x - j_1)$ satisfies $\alpha + j_1$. Since the substitution $x \mapsto x - j_1$ is an automorphism, $p(x - j_1)$ is irreducible, i.e., $p(x - j_1)$ is the minimal polynomial for $\alpha + j_1$ over $\mathbb{F}_p$. Hence

$$p(x - j_1) \mid f_1(x).$$

If $f_1(x)$ is irreducible, then $f_1(x) = p(x - j_1)$ and $f(x) = p(x)p(x - j_1)$ gives the factorizaiton of f. If $f_1(x)$ is not irreducible, then writing $f_1(x) = p(x - j_1)f_2(x)$, noticing f_2 contains a root $\alpha + j_2$ whose minimal polynomial is $p(x - j_2)$, etc. gives the irreducible factorization

$$f(x) = p(x)p(x - j_1)p(x - j_2) \cdots p(x - j_k)$$

for some k. Each irreducible factor is a substitution of $p(x)$, so each factor has the same degree as $p(x)$, giving

$$\deg p(x) \mid \deg f(x).$$

This contradicts the assumption that $1 < \deg p(x) < \deg f(x)$ and $\deg f(x) = p$, a prime.